

Preliminaries

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1 Installing the Required Tools

I use Arch Linux as my operating system. I use `pacman` to manage packages.

I found that the package names listed in the assignment correspond directly to packages in `pacman`. I used the `--needed` flag to avoid reinstalling packages if they were already present in my system.

Terminal

```
→ ricardo ~ sudo pacman -S --needed arm-none-eabi-gcc
→ ricardo ~ sudo pacman -S --needed arm-none-eabi-binutils
→ ricardo ~ sudo pacman -S --needed qemu-system-arm
```

Now, I ran the `-ql` flag on each of them to see which files those packages brought to my computer. I realized that the output was super long, so I redirected it to a file.

Terminal

```
→ ricardo ~ pacman -ql arm-none-eabi-gcc > arm-none-eabi-gcc-files.txt
→ ricardo ~ pacman -ql arm-none-eabi-binutils > arm-none-eabi-binutils-files.txt
→ ricardo ~ pacman -ql qemu-system-arm > qemu-system-arm-files.txt
```

For `arm-none-eabi-gcc`: It contains 2,969 entries. Using the folders, I grouped the contents like this:

1. The path `/usr/arm-none-eabi/include/c++/` contains $\sim 1,650$ entries ($\sim 56\%$) consisting mainly of C++ headers.
2. The path `/usr/arm-none-eabi/lib/` contains ~ 250 entries ($\sim 9\%$) related to the required libraries for the package.
3. The path `/usr/bin/` contains ~ 20 entries ($\sim 1\%$) related to the executables. I assume this because it is included in my `$PATH`.
4. The path `/usr/lib/gcc/arm-none-eabi/` contains $\sim 1,000$ entries (35%) related to GCC drivers.
5. The last path `/usr/share/man/` contains manual pages for the installed commands.

For **arm-none-eabi-binutils**: It contains 84 entries.

1. The path `/usr/arm-none-eabi/bin/` contains ~ 10 entries ($\sim 12\%$) consisting mainly of binaries.
2. The path `/usr/arm-none-eabi/lib/` contains ~ 20 entries ($\sim 24\%$) related to the required libraries for the package.
3. The path `/usr/bin/` contains ~ 15 entries ($\sim 18\%$) related to the executables.
4. The path `/usr/share/man/` contains the remaining files ($\sim 46\%$) for manual pages.

For **qemu-system-arm**: It contains only 16 entries.

1. The executable in the path `/usr/bin/`.
2. Some licenses in the path `/usr/share/licenses/`.
3. Manual pages at the path `/usr/share/man/`.

2 qemu-system-arm

Now, for this part of the assignment, I tried to run the command provided to run the “hello world,” but I got this message.

Terminal

```
→ ricardo homework1/problems qemu-arm cpu-emu-test
zsh: command not found: qemu-arm
```

Checking with my trusty LLM, I realized that I was missing the package **qemu-user**, so I installed it using **pacman** as I did before.

Terminal

```
→ ricardo homework1/problems sudo pacman -S --needed qemu-user
```

For **qemu-user**: It contains only 45 entries.

1. Some required tool executables in the path `/usr/bin/`.
2. Manual pages at the path `/usr/share/man/` as usual.

Now, it runs correctly!

Terminal

```
→ ricardo homework1/problems qemu-arm cpu-emu-test
Hello world!
```

Continuing with the assignment, I ran the command to emulate a machine:

Terminal

```
→ ricardo ~ qemu-system-arm -machine mcimx6ul-evk
VNC server running on ::1:5900
```

I was expecting a monitor interface as shown in the reading. I again asked my trusty LLM for options to solve this problem.

I have two options to address this:

1. To install `qemu-ui-gtk` that would provide my system with a GTK interface to the emulated machine.
2. To add the flags `-display none -monitor stdio` so it will directly open the monitor in my terminal emulator.

I chose option 2 because I prefer to avoid unnecessary packages. Now, my updated command works correctly:

Terminal

```
→ ricardo ~ qemu-system-arm -machine mcimx6ul-evk -display none -monitor stdio

QEMU 11.1.1 monitor - type 'help' for more information
(qemu)
```

Now the setup is completed!

3 Building ARM binaries

Let's save the file `babysteps.s` with the recommended formatting:

babysteps.s

```
1  .global _start
2
3  _start:
4  top:
5      eor r1, r1
6      b top
```

To build this, I ran the commands recommended in the readings:

Terminal

```
→ ricardo homework1/problems arm-none-eabi-as babysteps.s -o babysteps.o
→ ricardo homework1/problems arm-none-eabi-ld -g babysteps.o -o babysteps.elf
```

Investigating by myself, I was able to understand what those commands do:

- The first command `arm-none-eabi-as` builds an object file for ARM architecture. It reads the file `babysteps.s` and then the flag `-o` indicates the output path `babysteps.o`.
- The second command `arm-none-eabi-ld` links the object file assigning memory addresses and resolving symbols. It reads the file `babysteps.o` and then the flag `-o` indicates the output path `babysteps.elf`. The flag `-g` is just a compatibility flag; it is ignored.

Now, decomposing the assembly source code, I would describe it as follows:

- `.global _start` declares the symbol `_start` as global.
- `_start:` indicates that the symbol `_start` starts in that position.
- `top:` indicates that the symbol `top` starts in that position.
- `eor r1, r1` indicates an XOR operation. The evaluation of $r1 \oplus r1$ is assigned to `r1`.
- `b top` indicates a jump to the `top:` symbol.

I would summarize the assembly as the following “equivalent” `c` code that is more human-readable:

`babysteps.c`

```
1  int start(){
2      int r1;
3      while(1){
4          r1 = r1 ^ r1;
5      }
6  }
```

It contains an endless loop with an assignment of an XOR operation.

4 Studying the compiled binary

I ran the command `readelf` as shown in the reading.

Terminal

```
→ ricardo homework1/problems readelf -h babysteps.elf
ELF Header:
  Magic:   7f 45 4c 46 01 01 01 00 00 00 00 00 00 00 00
  Class:                               ELF32
  Data:                                   2's complement, little endian
  Version:                               1 (current)
  OS/ABI:                                UNIX - System V
  ABI Version:                           0
  Type:                                   EXEC (Executable file)
  Machine:                               ARM
  Version:                               0x1
```

```

Entry point address:      0x8000
Start of program headers: 52 (bytes into file)
Start of section headers: 4580 (bytes into file)
Flags:                   0x5000200, Version5 EABI, soft-float ABI
Size of this header:      52 (bytes)
Size of program headers:  32 (bytes)
Number of program headers: 1
Size of section headers:  40 (bytes)
Number of section headers: 8
Section header string table index: 7

```

After investigating documentation, the command `readelf` displays all the information related to ELF files. ELF files are usually executable files in Unix systems.

Summarizing the information there:

- **Magic:** Traditional file magic number. The bytes `7f 45 4c 46` correspond to the ELF magic number.
- **Class:** 32-bit ELF format.
- **Data:** Little-endian representation.
- **Version:** ELF format version.
- **OS/ABI:** Operating system and ABI for which the file was generated.
- **ABI Version:** Version of the specified ABI.
- **Type:** ELF file type. `EXEC` means that it is an executable file.
- **Machine:** Target processor architecture.
- **Version:** ELF object file version.
- **Entry point address:** Address where program execution begins.
- **Start of program headers:** Offset from the beginning of the file to the program header table.
- **Start of section headers:** Offset from the beginning of the file to the section header table.
- **Flags:** Architecture-specific flags.
- **Size of this header:** Size of the ELF header.
- **Size of program headers:** Size of each entry in the program header table.
- **Number of program headers:** Number of entries in the program header table.
- **Size of section headers:** Size of each entry in the section header table.
- **Number of section headers:** Number of entries in the section header table.

- **Section header string table index:** Specifies which section header contains the string table used to store section names.

I ran the command `readelf` with the flag `--section-headers` to get more information.

Terminal

```
→ ricardo homework1/problems readelf --section-headers babysteps.elf
There are 8 section headers, starting at offset 0x11e4:
```

Section Headers:

[Nr]	Name	Type	Addr	Off	Size	ES	Flg	Lk	Inf	Al
[0]		NULL	00000000	000000	000000	00		0	0	0
[1]	.text	PROGBITS	00008000	001000	000008	00	AX	0	0	4
[2]	.persistent	PROGBITS	00009008	001008	000000	00	WA	0	0	4
[3]	.noinit	NOBITS	00009008	000000	000000	00	WA	0	0	4
[4]	.ARM.attributes	ARM_ATTRIBUTES	00000000	001008	000012	00		0	0	1
[5]	.symtab	SYMTAB	00000000	00101c	000120	10		6	8	4
[6]	.strtab	STRTAB	00000000	00113c	000062	00		0	0	1
[7]	.shstrtab	STRTAB	00000000	00119e	000045	00		0	0	1

Key to Flags:

W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
 L (link order), O (extra OS processing required), G (group), T (TLS),
 C (compressed), x (unknown), o (OS specific), E (exclude),
 D (mbind), y (purecode), p (processor specific)

Again, I investigated about the information here and I summarize it:

- **Index 0:** Empty by convention.
- **.text:** Executable instructions of the program. It is allocated in memory and executable.
- **.persistent:** Data intended to persist across certain executions or resets. It is writable and allocated.
- **.noinit:** Uninitialized data that requires memory space but does not need to be stored in the ELF file. It is writable and allocated.
- **.ARM.attributes:** ARM-specific metadata describing architecture and ABI attributes.
- **.symtab:** The symbol table contains information about functions and variables.
- **.strtab:** String table containing the names of functions and variables.
- **.shstrtab:** The string table with the names of the sections.

5 Disassembling the compiled binary

Let's disassemble the ELF file we got.

Terminal

```
→ ricardo homework1/problems arm-none-eabi-objdump -D babysteps.elf
```

```
babysteps.elf:      file format elf32-littlearm
```

```
Disassembly of section .text:
```

```
00008000 <_start>:
```

```
8000:      e0211001      eor      r1, r1, r1
8004:      eaffffff      b        8000 <_start>
```

```
Disassembly of section .ARM.attributes:
```

```
00000000 <.ARM.attributes>:
```

```
0: 00001141      andeq   r1, r0, r1, asr #2
4: 61656100      cmnvs   r5, r0, lsl #2
8: 01006962      tsteq   r0, r2, ror #18
c: 00000007      andeq   r0, r0, r7
10: Address 0x10 is out of bounds.
```

According to the output we got, I see in the `.text` section the code disassembled. The output includes memory addresses, but its logic is the same as the source code we had before.

Cleaning up the output of the disassembling command, I would recover the data like this:

babysteps_recovered.s

```
1  _start:
2      eor      r1, r1, r1
3      b        _start
```

I noticed the following points and I tried to give the best explanation I could:

1. **.global keyword is gone:** Investigating a little bit about this, the way to discover if a label is global is checking the symbol table. Using the command `readelf -s babysteps.elf`, I was able to see it as global.
2. **Label top: is gone:** In my experience working in compilers, it should've been removed by some sort of compiler optimization. It is clear that this label was not necessary due to the existence of top in the same memory address.
3. **eor instruction now has 3 parameters:** This is because the original `eor r1, r1` is just a shortcut of the 3-parameter version because the destination register is the same as the first operand of the XOR operation.

Essentially, the changes are for optimization, but the semantics of the sources are equivalent.

6 Running ARM binaries we just built

To run the binaries in an emulated machine, I crafted the following command.

Terminal

```
→ ricardo homework1/problems qemu-system-arm
  -machine mcimx6ul-evk
  -kernel babysteps.elf
  -display none
  -monitor stdio
QEMU 11.1.1 monitor - type 'help' for more information
(qemu) stop
(qemu) info registers

CPU#0
R00=00000000 R01=00000000 R02=00000000 R03=00000000
R04=00000000 R05=00000000 R06=00000000 R07=00000000
R08=00000000 R09=00000000 R10=00000000 R11=00000000
R12=00000000 R13=00000000 R14=00000000 R15=00008000
...
FPSCR: 00000000
(qemu) quit
```

I removed part of the output to keep only the important information; I put ... where I cut the info.

The program didn't print any info (because it is not supposed to do so), but I can see that the register **R15=00008000** is moving in the endless loop.

To run this, I had to:

- Install the required packages in my Arch Linux installation using pacman.
 - The required compiler extensions.
 - Utils for binaries.
 - QEMU as the full hardware emulator (`qemu-system-arm`).
- Build a sample assembly `babysteps.s` and then run it in a full hardware emulation of the board `mcimx6ul-evk`.

Now, I investigated (with a little bit of help of my agent) which board in the catalog of emulable boards is the closest to the `USB Armory MKII`.

I found the board `mcimx6ul-evk` as the closest for the following reasons:

- Both use processors from the `NXP i.MX6UL` family.
- Both use an `ARM Cortex-A7 CPU` core.
- The emulated board has peripherals and a memory layout similar to those of an embedded `i.MX6UL` system.

I will continue using this `mcimx6ul-evk` board from now on.

i Extra Credit Modify the assembly code to print a string to the serial0 port upon startup.

For this, I modified the file as follows:

 `babysteps_serial.s`

```
1  .equ UART1_UTXD, 0x02020040
2
3  .section .text
4  .global _start
5
6  _start:
7      ldr r0, =message
8      ldr r1, =UART1_UTXD
9
10     print_character:
11         ldrb r2, [r0], #1
12         cmp r2, #0
13         beq finished
14         str r2, [r1]
15         b print_character
16
17     finished:
18         b finished
19
20     message:
21         .asciz "Hello from serial0!\r\n"
```